The construction of four-dimensional quantum Yang–Mills theory

Under construction. Version of November 11, 2026: this manuscript is incomplete. Sections are added and revised as they are written and checked against the formal proof; see the status table at https://github.com/scottnarmstrong/yang-mills-informalization/tree/main/manuscript .

Abstract

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Part I
The theorem and the architecture of its proof

1 Introduction

2 Wilson theory

The starting measure assigns a group element to each lattice link and weights the curvature around each plaquette. We first fix the matrix normalization: it determines both the Wilson energy and the coefficient of the logarithmic coupling flow.

Throughout, $\mathbb{N} = \{0, 1, \dots\}$, coordinate indices run from 1 to 4, and e_a is the ath coordinate vector. Time is the fourth coordinate. For the later relativistic statements write

$$\mathsf{m}(x) = (x^4)^2 - |\vec{x}|^2, \qquad \overline{V}_+ = \{\mathbf{p} : |\vec{\mathbf{p}}| \le \mathbf{p}^4\}\,.$$

Inner products are antilinear in the first variable, and spacetime integrals use Lebesgue measure. Arity zero is included; empty sums are zero and empty products are one. The group H_4 consists of the signed permutation matrices; $\mathsf{s}_a^{\mathrm{refl}}$ denotes reflection in coordinate a.

Definition 2.1 (Gauge normalization)**.** Fix $n_G \ge 1$ and a closed subgroup $G \subset U(n_G)$ with simple nonabelian real Lie algebra

$$\mathfrak{g} = \{X \in M_{n_G}(\mathbb{C}) : \exp(tX) \in G \text{ for every } t \in \mathbb{R}\}, \qquad \langle X, Y\rangle_{\mathfrak{g}} = -n_G^{-1}\operatorname{Re}\operatorname{tr}(XY)\,.$$

For an orthonormal basis $(e_\alpha^{\mathfrak{g}})$, put

$$d_{\mathfrak{g}} = \dim_{\mathbb{R}} \mathfrak{g}, \qquad C_{\mathfrak{g}} = -d_{\mathfrak{g}}^{-1}\sum_\alpha \operatorname{tr}_{\mathfrak{g}}\big(\operatorname{ad}(e_\alpha^{\mathfrak{g}})^2\big),$$

$$b_0 = \frac{11 C_{\mathfrak{g}}}{48\pi^2}, \qquad \kappa_G = 2b_0\,.$$

This fixes the normalization of the running coupling.

Here $\operatorname{ad}(X)Y = [X, Y]$ and $\operatorname{tr}_{\mathfrak{g}}$ is the real operator trace on $\mathfrak{g}$. Simplicity means $[\mathfrak{g}, \mathfrak{g}] \neq 0$ and no proper nonzero real ideal. Connectedness of G is not assumed. The adjoint maps are skew-adjoint for this metric; nonabelianity therefore gives $C_{\mathfrak{g}} > 0$ and $\kappa_G > 0$. For $u \in G$, set

$$\operatorname{pr}(u) = 1 - n_G^{-1}\operatorname{Re}\operatorname{tr} u, \qquad \operatorname{ch}(u) = n_G^{-1/2}\|u - 1\|_{\mathrm{F}}, \qquad \operatorname{ch}(u)^2 = 2\operatorname{pr}(u)\,. \tag{2.1}$$

The first quantity is the plaquette energy; the second measures its distance from the identity. Both are invariant under conjugation and inversion.

Definition 2.2 (Wilson measure)**.** For an integer $s \ge 1$ and $\beta \in \mathbb{R}$, put $\Lambda_s = (\mathbb{Z}/s\mathbb{Z})^4$. A configuration assigns $U(y, a) \in G$ to every positive link, with inverse on the reversed link. Define

$$U_{y;ab} = U(y, a)U(y + e_a, b)U(y + e_b, a)^{-1}U(y, b)^{-1},$$

$$E_s^{\mathrm{W}}(U) = \sum_{y\in\Lambda_s}\sum_{a<b}\operatorname{pr}(U_{y;ab}),$$

$$d\mu_{s,\beta}(U) = Z_{s,\beta}^{-1}e^{-\beta E_s^{\mathrm{W}}(U)}\prod_{y,a} dh(U(y, a))\,.$$

Here dh is probability Haar measure and $Z_{s,\beta}$ normalizes the measure. This is the lattice theory to be scaled.

The normalization is the positive finite integral

$$Z_{s,\beta} = \int e^{-\beta E_s^{\mathrm{W}}(U)} \prod_{y,a} dh(U(y,a)) \,.$$

Indeed, the integrand is continuous and strictly positive on the compact configuration space. A site gauge transformation $\gamma_{\mathrm{g}} : \Lambda_s \to G$ acts by

$$U^{\gamma_{\mathrm{g}}}(y,a) = \gamma_{\mathrm{g}}(y) U(y,a) \gamma_{\mathrm{g}}(y+e_a)^{-1} \,.$$

It changes each plaquette by conjugation at its base point, so the energy and the Wilson measure are gauge invariant. Lattice isometries transport links and invert reversed ones. The measure is invariant under these transformations as well; the finite-lattice identities are proved in Section 9.

A general Wilson family consists of strictly increasing positive integer sequences (L_p) and (D_N), real couplings $\beta_{N,p}$, a label set $\mathcal{T}$, and complex insertions $I_{N,p}(v,y;U)$. The index N controls the spacing, and p controls the physical volume. Set

$$a_N = D_N^{-1}, \qquad \mu_{N,p} = \mu_{L_p D_N, \beta_{N,p}}, \qquad \hat{y} = a_N \big(r_{L_p D_N}(y^a)\big)_{a=1}^4 \,, \tag{2.2}$$

where $r_s(j)$ is the integer representative of $j \in \mathbb{Z}/s\mathbb{Z}$ in $(-s/2, s/2]$. Thus the physical torus has side L_p.

Definition 2.3 (Lattice moments)**.** For a label list $v \in \mathcal{T}^n$, sites $y \in \Lambda_{L_p D_N}^n$, and a test $f : (\mathbb{R}^4)^n \to \mathbb{C}$, define

$$M_{N,p}^{\mathrm{lat}}(v;y) = \int \prod_{i=1}^n I_{N,p}(v_i, y_i; U)\, d\mu_{N,p}(U),$$
$$C_{N,p}^{\mathrm{lat}}(v;f) = a_N^{4n} \sum_{y \in \Lambda_{L_p D_N}^n} f(\hat{y}) M_{N,p}^{\mathrm{lat}}(v;y) \,.$$

These are the point correlations and their physical Riemann sums.

The construction below chooses the spacings, couplings and local insertions together. At each fixed physical volume the ultraviolet limit will use the full sequence $N \to \infty$; only the subsequent infinite-volume limit requires a subsequence.

3 Renormalization

Blocking integrates one scale against a probability kernel. A curvature action and two activity families describe the new density. The inductive step controls these coordinates and the coupling; the multiscale theorem chooses the bare coupling to reach one prescribed endpoint.

Fix $\mathsf{b} > 0$. Throughout we use the common choices of Sections 9–13, made once for this group realization and penalty in the order of Section A. Their existence is proved with Propositions 3.5 and 3.6. Constants may depend on these choices; further dependence is stated where it occurs.

3.1 Scales and the block map

Levels decrease under blocking. With odd ratios $\ell_m \geq 3$, set

$$D_0 = 1, \qquad D_{m+1} = \ell_m D_m, \qquad a_m = D_m^{-1}, \qquad s_m = \log D_m \,. \tag{3.1}$$

The level-m torus has side L_pD_m in lattice units and side L_p in physical units. Starting at cutoff N, the number of completed steps is $k = N - m$; the step $m+1 \to m$ advances this induction from k to $k+1$.

The stage J_m records the available derivative order and bounds. The stages satisfy $4 \le J_m \le J_{m+1} \le J_m + 1$ and $J_m \to \infty$. Fix the reference endpoint $\bar\beta > 1$ and define

$$\widehat\beta_m = \bar\beta + \kappa_G s_m, \qquad M_m^{\mathrm{cut}} = \exp\big((\log\log\widehat\beta_m)^2\big), \qquad t_m = \widehat\beta_m^{-1/2} M_m^{\mathrm{cut}}, \tag{3.2}$$

$$B_m = \lceil (\log\widehat\beta_m)^2 \rceil, \qquad \epsilon_m^{\mathrm{reg}} = \widehat\beta_m^{-11/10}, \qquad \epsilon_m^{\mathrm{for}} = e^{-(M_m^{\mathrm{cut}})^{1/5}}. \tag{3.3}$$

The reference coupling $\widehat\beta_m$ tracks the leading running slope. The factor M_m^{cut} enlarges the Gaussian field scale; t_m is the threshold separating small and large plaquette fields. The radius B_m sets the scale of the support weights. The quantities $\epsilon_m^{\mathrm{reg}}$ and $\epsilon_m^{\mathrm{for}}$ bound the regular and forced activity sums, respectively. These thresholds depend on the reference coupling, so they remain fixed as the actual coupling varies.

The allowed coupling interval is

$$\mathfrak{B}_m = [\widehat\beta_m/2, 2\widehat\beta_m]\,.$$

At stage j, write $\mathfrak{q}_j$ for the derivative order, ϖ_j for the spatial power, σ_j for the support exponent and $\omega_j^{\mathrm{RG}} \ge 0$ for the coupling drift bound. Their choices and growth conditions are in Section A.

At a step $m+1 \to m$, the fine links U and coarse links V lie on $\Lambda_{L_pD_{m+1}}$ and $\Lambda_{L_pD_m}$. Put $j = J_{m+1}$ and $\ell = \ell_m$. On patches of radius $(j+3)\ell$, symmetric frames express fine straight connectors in the patch frames. Their averages, with the alternative connector choices needed outside the minimizing branch, give group-valued predictors $Q_e^{\mathrm{pred}}(U, \alpha_{\mathrm{tag}})$. The finite tag list α_{tag} specifies these frame and predictor choices. Its product probability mass is $\lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}})$; the exact frame and predictor formulas are given in Section 10, Definitions 10.1–10.2.

Definition 3.1 (Exact block map)**.** For fine links U and coarse links V, define

$$k_\beta^{(m)}(U,V) = \sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U,\alpha_{\mathrm{tag}}) \prod_{e\ \mathrm{coarse}} \frac{e^{-\mathsf{b}\beta\,\mathrm{pr}(V_e Q_e^{\mathrm{pred}}(U,\alpha_{\mathrm{tag}})^{-1})}}{\int_G e^{-\mathsf{b}\beta\,\mathrm{pr}(u)}\,dh(u)},$$

$$(\mathcal{B}_\beta^{(m)} f)(V) = \int k_\beta^{(m)}(U,V) f(U)\,dU\,.$$

Both product measures are probability Haar measures, and $\int k_\beta^{(m)}(U,V)\,dV = 1$. The map integrates one scale without changing the partition integral.

For fixed U and tags, right invariance of Haar measure makes every coarse-link factor integrate to one. Summing the tag masses proves the normalization. Consequently, for integrable f,

$$\int (\mathcal{B}_\beta^{(m)} f)(V)\,dV = \int f(U)\,dU\,. \tag{3.4}$$

The branch choices, measurability and symmetry properties of this kernel are proved in Section 10.

3.2 The density and its norms

At level m, write $\Lambda = \Lambda_{L_pD_m}$. Let $\mathrm{Flag}_m(U)$ be the sites incident to a plaquette whose chord exceeds t_m. A support is healthy if it avoids this set. A regular activity E_i^{reg} has a site support X_i

and an anchor $o_i \in X_i$. A forced activity K_i^{for} has a support Y_i and a nonempty inventory $I_i \subset Y_i$ of flagged sites. Its value vanishes unless $Y_i \cap \mathrm{Flag}_m(U) = I_i$ and its fixed local extra test holds. The activity families are countable.

The scalar v_{vac} records the vacuum term. The fixed action $P_{N,m}^{\mathrm{path}}(\beta, U)$ is the clipped curvature action of Section 11, Definitions 11.1–11.2; its path arrays are independent of p. It retains the quadratic, cubic and quartic curvature terms, with β the running kinetic coefficient. Write $\beta(\rho)$ for this coefficient in a representation ρ.

Definition 3.2 (Effective density)**.** For regular activities $(X_i, o_i, E_i^{\mathrm{reg}})$ and forced activities $(Y_i, I_i, K_i^{\mathrm{for}})$, define

$$\operatorname{dens}\rho(U) = e^{v_{\mathrm{vac}}|\Lambda| - P_{N,m}^{\mathrm{path}}(\beta,U) - \sum_i \mathbf{1}_{X_i \text{ healthy}} E_i^{\mathrm{reg}}(U)} \cdot \sum_{\substack{\mathcal{A}_{\mathrm{for}} \subset \mathbb{N} \text{ finite} \\ Y_i \ (i\in\mathcal{A}_{\mathrm{for}}) \text{ pairwise disjoint} \\ \bigcup_{i\in\mathcal{A}_{\mathrm{for}}} I_i = \mathrm{Flag}_m(U)}} \prod_{i\in\mathcal{A}_{\mathrm{for}}} K_i^{\mathrm{for}}(U)\,.$$

The sum covers every flagged site; each inventory is a nonempty subset of its support. This represents the density after integration.

If there are no flagged sites, only the empty family occurs and the forced sum equals one. When flags are present, their exact coverage and the disjointness of the supports assign each flagged site to precisely one selected forced term. Regular activities enter only on healthy supports. Their local regularity and cancellations, and the restrictions on forced activities, are specified in Section 12, Definitions 12.1–12.2.

Distances below use the unit-lattice sup metric. For a link e, let e^- be its initial site. The left derivative $D_{\dot{\mathcal{A}}}^L$ differentiates the increments $U_e \mapsto e^{t\dot{\mathcal{A}}_e} U_e$, with $\dot{\mathcal{A}}_e \in \mathfrak{g}$ and norm from Definition 2.1.

Definition 3.3 (Local derivative seminorm)**.** For a regular activity anchored at o_i, set

$$\|E_i^{\mathrm{reg}}\|_m = \sup_{0\le q\le \mathfrak{q}_{J_m}} \sup_{U,\dot{\mathcal{A}}_1,\dots,\dot{\mathcal{A}}_q} \left| D_{\dot{\mathcal{A}}_1}^L \cdots D_{\dot{\mathcal{A}}_q}^L E_i^{\mathrm{reg}}(U) \right|.$$

The second supremum is over configurations healthy on X_i and directions satisfying

$$|(\dot{\mathcal{A}}_h)_e| \le t_m \big(1 + \mathrm{dist}(e^-, o_i)\big)^{\varpi_{J_m}}.$$

This measures ordered local derivatives in units of the permitted field size.

For a support X, let $\mathrm{cov}_{B_m}(X)$ be the minimum number of radius-B_m lattice balls with connected union covering X. The exponential weight below penalizes supports that require many such balls.

Definition 3.4 (Weighted activity sums)**.** For nonnegative majorants $e_i^{\mathrm{maj}} \ge \|E_i^{\mathrm{reg}}\|_m$ and $k_i^{\mathrm{maj}} \ge \sup_U |K_i^{\mathrm{for}}(U)|$, define

$$\mathcal{E}_m = \sup_x \sum_{o_i = x} e^{\sigma_{J_m} \mathrm{cov}_{B_m}(X_i)} e_i^{\mathrm{maj}},$$
$$\mathcal{K}_m = \sup_x \sum_{x\in Y_i} e^{\sigma_{J_m} \mathrm{cov}_{B_m}(Y_i)} k_i^{\mathrm{maj}}.$$

These sums measure regular and forced terms separately, with a penalty for large supports.

The chosen admissible classes $\mathcal{A}_{N,m,p}$, initial classes $\mathcal{I}_{N,p} \subset \mathcal{A}_{N,N,p}$, and adjacent-level relations

$$\mathcal{R}_{N,m,p} \subset \mathcal{A}_{N,m,p} \times \mathcal{A}_{N,m+1,p}$$

are constructed in Section 12, D.2. Their members have the density of Definition 3.2, and their membership retains the continuation conditions of that construction. For every member, D.1 supplies majorants with convergent weighted sums satisfying

$$\mathcal{E}_m \le \epsilon_m^{\mathrm{reg}}, \qquad \mathcal{K}_m \le \epsilon_m^{\mathrm{for}}\,. \tag{3.5}$$

These inequalities are properties of the chosen classes; membership also requires the retained continuation conditions.

Along a continuous curve, supports, anchors, inventories and extra tests are fixed, and the coupling and vacuum scalar are continuous. Differences of regular derivatives on healthy configurations and of forced values admit majorants whose weighted sums over all addresses tend to zero as the parameter approaches any point. This is the continuity of Definition 12.3 in Section 12; flag boundaries are not differentiated.

3.3 The inductive step

Exact integration removes one scale. Extracting its marginal curvature-square term into the coupling leaves a bounded drift error and returns a density in the next admissible class.

Proposition 3.5 (One step)**.** *For every N, p, $m < N$, compact real interval $[u_0, u_1]$ and continuous $\rho'_u \in \mathcal{A}_{N,m+1,p}$ with $\beta'_u = \beta(\rho'_u) \in \mathfrak{B}_{m+1}$, there is a continuous $\rho_u \in \mathcal{A}_{N,m,p}$ with $(\rho_u, \rho'_u) \in \mathcal{R}_{N,m,p}$ and*

$$\begin{aligned} \operatorname{dens}\rho_u &= \mathcal{B}^{(m)}_{\beta'_u} \operatorname{dens}\rho'_u, \\ \beta(\rho_u) &= \beta'_u - \kappa_G \log \ell_m + e_u, \qquad |e_u| \le \omega^{\mathrm{RG}}_{J_{m+1}}\,. \end{aligned}$$

The proof occupies Sections 10–12, ending with D.3. Symmetric gauge fixing and the stationary background give the small-field integration; the large-field lower bound controls the forced terms. Fourth-order Wilson matching and the scalar coupling reset identify the displayed drift in Section 11. Section 12 assembles the activities with the bounds (3.5).

The constants ω^{RG}_j depend on the stage and the fixed construction. RG imposes a band condition only on the input coupling. The trajectories of MT below also have their output couplings in the appropriate bands. Section 12, D.3 supplies the continuous initial curve

$$\beta \longmapsto \rho^{\mathrm{W}}_\beta \in \mathcal{I}_{N,p}, \qquad \beta(\rho^{\mathrm{W}}_\beta) = \beta, \qquad \operatorname{dens}\rho^{\mathrm{W}}_\beta = e^{-\beta E^{\mathrm{W}}_{L_p D_N}} \quad (\beta \in \mathfrak{B}_N)\,, \tag{3.6}$$

whose vacuum scalar is zero. This is the base case for the induction.

3.4 Multiscale trajectories

A trajectory is a tuple $\rho_{N,p,m} \in \mathcal{A}_{N,m,p}$, $0 \le m \le N$, satisfying

$$\rho_{N,p,N} \in \mathcal{I}_{N,p}, \qquad (\rho_{N,p,m}, \rho_{N,p,m+1}) \in \mathcal{R}_{N,m,p} \quad (m < N)\,.$$

Write $\beta_{N,p,m} = \beta(\rho_{N,p,m})$. Shooting through the coupling bands selects a bare coupling whose trajectory ends at the common value $\bar\beta$. This fixes the bare family before any volume sequence is prescribed.

Proposition 3.6 (Multiscale trajectory)**.** *For every N, p, there are a bare coupling $\beta_{N,p}$ and a trajectory with $\rho_{N,p,N} = \rho^{\mathrm{W}}_{\beta_{N,p}} \in \mathcal{I}_{N,p}$ such that*

$$\beta_{N,p,0} = \bar{\beta}, \qquad \beta_{N,p,m} \in \mathfrak{B}_m \quad (0 \le m \le N),$$
$$\operatorname{dens} \rho_{N,p,m} = \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} \operatorname{dens} \rho_{N,p,m+1} \quad (m < N).$$

For every $\varepsilon > 0$, some $m_0(\varepsilon)$ satisfies

$$|\beta_{N,p,m} - \kappa_G s_m| \le \varepsilon s_m \qquad (N \ge m \ge m_0,\ p \in \mathbb{N}).$$

The proof is in Section 13, E.1.1–E.1.2. The exact drift bound of RG holds on every adjacent pair, and all trajectory integrals equal $Z_{L_p D_N, \beta_{N,p}}$ by (3.4) and the Wilson start. The path arrays remain independent of p.

The running law follows by summing the drift from the fixed endpoint:

$$|\beta_{N,p,m} - \widehat{\beta}_m| \le \sum_{i=0}^{m-1} \omega^{\mathrm{RG}}_{J_{i+1}}. \tag{3.7}$$

The choices in Section A give $\omega^{\mathrm{RG}}_{J_{i+1}} / \log \ell_i \le J_{i+1}^{-2}$. Since $J_i \to \infty$ and $s_m = \sum_{i<m} \log \ell_i \to \infty$, the right side of (3.7) is $o(s_m)$: split the sum at a fixed level, then bound the tail by $\sup_{i \ge i_0} J_{i+1}^{-2}$ times s_m. The finite initial sum and $\bar{\beta}$ disappear after division by s_m, uniformly in $N \ge m$ and p.

The same level choices give $\log \ell_m \to \infty$ and $\log \ell_m / s_m \to 0$. The physical periods L_p are strictly increasing multiples of four, and every positive integer divides L_p for infinitely many p. Analytic endpoint tuning, used for the full ultraviolet limit, is proved in Section 13, E.1.3.

3.5 Local fields

Local fields are normalized along the same trajectories. Their polynomial labels specify the continuum Taylor coefficients, while transfer to a fixed reference level determines the insertion at the fine cutoff.

Definition 3.7 (Polynomial labels)**.** Let $\mathcal{J}$ consist of formal $\mathfrak{g}$-valued arrays $\mathcal{A}_a(x) = \sum_{k \ge 0} \mathcal{A}_{k,a}(x)$, with $\mathcal{A}_{k,a}$ homogeneous of degree $k+1$ and

$$\sum_a x^a \mathcal{A}_{k,a}(x) = 0, \qquad V_d = \mathbb{C}[\mathcal{J}]^{\mathrm{Ad}(G)}_{\mathrm{weight} \le d}.$$

Each coordinate of $\mathcal{A}_k$ has bidegree $(k+2, 1)$: engineering weight and number of polynomial factors. Fix bihomogeneous bases, the supremum coefficient norm, conjugation and the induced H_4 action. These labels describe invariant curvature polynomials; curvature has weight two and a covariant derivative adds one.

Write $\mathbf{1}$ for the constant polynomial one. Section 9, A.1 constructs a local section $P^{\mathrm{loc}}_d[\mathbf{c}](y; U)$, linear in $\mathbf{c} \in V_d$ and polynomial in the real and imaginary parts of link matrix entries on a fixed finite stencil $\Sigma_d \subset \mathbb{Z}^4$. It is gauge invariant, commutes with conjugation, is H_4 covariant, and sends the polynomial unit to one. Its Taylor normalization is stated there.

Choose the reference level $m_{\mathrm{ref}}(d)$ with $J_{m_{\mathrm{ref}}(d)} \ge d{+}3$ and $\mathfrak{q}_{J_m} \ge d{+}1$, $\varpi_{J_m} \ge d$ for $m \ge m_{\mathrm{ref}}(d)$, as in Section 13, E.2.1. For a basis vector $b_{d,i}$ of bidegree (w_i, k_i) define the factor-count scaling by

$$\mathsf{D}^{\mathrm{fac}}_t b_{d,i} = t^{k_i} b_{d,i}.$$

For $N \geq m_{\mathrm{ref}}(d)$ and terminal input $\mathbf{v} \in V_d$, the marked density at the origin is initialized and evolved by

$$H^{\mathrm{ins}}_{N,p,N;d}(\mathbf{v};U) = \operatorname{dens} \rho_{N,p,N}(U)\, P^{\mathrm{loc}}_d[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}}\mathbf{v}](0;U), \tag{3.8}$$

$$H^{\mathrm{ins}}_{N,p,m;d}(\mathbf{v}) = \ell_m^d \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} H^{\mathrm{ins}}_{N,p,m+1;d}(\mathbf{v}) \quad (m_{\mathrm{ref}}(d) \leq m < N)\,. \tag{3.9}$$

Its representation by a local section and regular and forced remainders gives a coefficient vector $\mathbf{x}_m(\mathbf{v})$. The transfer construction in Section 13, E.2.1 supplies the invertible terminal map

$$\mathbf{x}_N(\mathbf{v}) = \mathbf{v}, \qquad \mathbf{x}_{m_{\mathrm{ref}}(d)}(\mathbf{v}) = \mathsf{Z}_{N,p,d}\mathbf{v}\,.$$

For $N < m_{\mathrm{ref}}(d)$ choose invertible maps to define the insertions; transfer normalization is required only at and above the reference level.

Definition 3.8 (Renormalized insertions)**.** Put $\mathcal{T} = \bigsqcup_{d\geq 0} V_d$ and define, at every cutoff,

$$I_{N,p}((d,\mathbf{c}),y;U) = O^{(d)}_{N,p}(\mathbf{c},y;U) = a_N^{-d} P^{\mathrm{loc}}_d[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}}\mathsf{Z}^{-1}_{N,p,d}\mathbf{c}](y;U)\,.$$

This fixes the insertion along the trajectory. Its distinguished unit coefficient is $\mathbf{u}_d = a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$, whose insertion equals one for $N \geq m_{\mathrm{ref}}(d)$. The scaling $\mathsf{D}^{\mathrm{fac}}_t$ counts factors, not engineering weight.

The inverse terminal map makes the transferred coefficient equal to the prescribed $\mathbf{c}$ at the reference level. The unit identity follows from $\mathbf{x}_m(a_N^d\mathbf{1}) = a_m^d\mathbf{1}$, proved with the transfer symmetries in Section 13, E.2.1a. In particular the unit coefficient retains the reference-spacing factor even in a degree-labelled space containing the constant polynomial.

The bare couplings of MT and these insertions now specify one Wilson family. Each insertion is continuous, gauge invariant and local on its fixed stencil. MT at $m = N$ gives the volume-uniform asymptotic slope $\kappa_G > 0$. These are the lattice inputs to the continuum limits.

4 The continuum limit

We first remove the lattice spacing at each fixed volume, then let the volume increase along one refinement. Throughout, the Wilson family and insertions are those of Section 3; they are fixed before any volume sequence is prescribed. Constants may depend on the fixed group, its realization, the penalty and the construction choices of Appendix A. Additional dependence is stated below.

For positions in $(\mathbb{R}^4)^n$, set

$$\mathrm{Conf}_n = \{x : x_i \neq x_j \text{ for } i \neq j\}, \qquad \mathrm{Gen}_n = \{x : x_i^a \neq x_j^a \text{ for } i \neq j,\ a = 1,\dots,4\},$$
$$\mathrm{NoHit}_{L,n} = \{x : x_i - x_j \notin L\mathbb{Z}^4 \text{ for } i \neq j\}\,.$$

The last domain excludes collisions on a torus of physical side L. A separated test is a smooth function on $(\mathbb{R}^4)^n$ with compact closed support in Conf_n. It is p-admissible if its support lies in $((-L_p/2, L_p/2)^4)^n$. Write $\mathrm{sep}_\infty(x) = \min_{i\neq j} |x_i - x_j|_\infty$, with value $+\infty$ for $n \leq 1$.

For a finite list $\boldsymbol{l}$ of slot–axis pairs, define

$$\Delta_{\boldsymbol{l}} = \prod_{(i,a)\in\boldsymbol{l}} a_N^{-1}(\tau_{i,a} - 1), \qquad \partial_{\boldsymbol{l}} = \prod_{(i,a)\in\boldsymbol{l}} \partial_{x_i^a}\,.$$

Here $\tau_{i,a}$ shifts slot i forward by one lattice link. These differences measure derivatives in physical units. Torus separation is the minimum, over distinct slots, of $\min_{z\in L_p\mathbb{Z}^4} |\hat{y}_i - \hat{y}_j + z|_\infty$.

The local estimates give bounds independent of the volume. They control compactness for every fixed list of fields and derivatives.

Proposition 4.1 (Uniform difference bounds)**.** *For every arity n, token list v, finite difference list $\boldsymbol{l}$ and $r > 0$, there are $B(n, v, \boldsymbol{l}, r) \geq 0$ and $N_0(n, v, \boldsymbol{l}, r)$ such that, for every $N \geq N_0$, every p and every lattice tuple of torus separation at least r,*

$$|\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}(v; y)| \leq B\,.$$

The proof is in Section 13, II.E.2.2–E.2.4. Marked transfers control insertion coefficients and remainders; the centered chessboard estimate then bounds correlations. An intermediate scale below the prescribed separation also permits finite differences. The chessboard constant in II.E.2.3 precedes that scale, although its cutoff threshold may depend on it. UB's bound and cutoff threshold are independent of p.

Analytic tuning to the common endpoint gives convergence along the full ultraviolet sequence. At a fixed volume, the limiting kernel is smooth even where two positions share some coordinates.

Theorem 4.2 (Fixed-volume limit)**.** *For every p, n, v, there is $f_{p,v} \in C^\infty(\mathrm{NoHit}_{L_p,n})$ such that, for every finite $\boldsymbol{l}$ and every p-admissible separated test f,*

$$\lim_{N\to\infty} a_N^{4n} \sum_y f(\hat{y}) \Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}(v; y) = \int f(x) \partial_{\boldsymbol{l}} f_{p,v}(x)\, dx\,.$$

The proof is in Section 13, II.E.3.1–E.3.2. At a fixed intermediate level, the normalized correlations are Cauchy as the cutoff increases. On common lattice configurations, comparison with the original correlations has an error tending to zero exponentially in B_m. First choose m, then the cutoffs. UB extends the resulting convergence from a dense set and identifies its derivatives. The proof also gives convergence at moving interpolated points on the periodic domain and integrability of the displayed integrand. Its thresholds may depend on p.

The following definition keeps the order of limits explicit. Write $C^\infty_p(v; f)$ for a fixed-volume smeared limit.

Definition 4.3 (Ordered convergence)**.** A Wilson family converges to K along v if, for every n, v and separated test f, the test is eventually $\mathsf{v}(k)$-admissible and there are numbers $C^\infty_p(v; f)$ with

$$\begin{aligned} &C^{\mathrm{lat}}_{N,p}(v; f) \longrightarrow C^\infty_p(v; f) \quad (N \to \infty) && \text{at every admissible } p,\\ &C^\infty_{\mathsf{v}(k)}(v; f) \longrightarrow \int f(x) K_n(v; x)\, dx\,. \end{aligned}$$

Values at inadmissible volumes are unrestricted.

We specify the Euclidean conditions for the multilinear table S to be constructed. Units, conjugation and the signed permutation action are inherited from the polynomial labels.

Definition 4.4 (E1–E3. Euclidean symmetries)**.** E1 is simultaneous permutation symmetry, $S_0 = 1$ and deletion of a unit slot. E2 is

$$S_n(hc; hx) = S_n(c; x) \quad (h \in H_4), \qquad S_n(\bar{c}; x) = \overline{S_n(c; x)}\,.$$

E3 is translation invariance.

A word is a finite list of labels at separated positions; it is positive-side for axis a when all its positions have positive ath coordinate. The product of words concatenates their lists, so reflection positivity tests a correlation with one reflected group of insertions.

Definition 4.5 (E4–E5. Reflection positivity)**.** E4 requires, for every axis a, positive-side separated words w_i and complex coefficients z_i,

$$\sum_{i,j} \bar{z}_i z_j S(\theta_a w_i^* w_j) \geq 0 \,.$$

The star reverses the word and applies conjugation to its labels; θ_a reflects positions and acts on labels. E5 requires the same positivity for finite sums of words smeared against joint smooth compactly supported tests on positive-side separated configurations, and absolute integrability of every mixed pairing.

For a fixed label list, D_x^k denotes the kth Fréchet derivative in the positions, with the product Euclidean norm. The next bound controls approach to collision sets uniformly over translations and over configurations with large diameter.

Definition 4.6 (E6–E7. Euclidean regularity)**.** E6 is smoothness on Conf_n. E7 requires, for each n, d_i, c_i, k, constants $C(n,d,c,k) \geq 0$ and an integer $A(n,d,c,k) \geq 0$ such that

$$\|D_x^k S_n(c;x)\| \leq C r^{-A} \qquad (0 < r \leq 1,\ \mathrm{sep}_\infty(x) \geq r) \,.$$

One volume refinement will serve every field and all subsequent constructions.

Theorem 4.7 (Infinite-volume limit)**.** *For every strictly increasing* $\mathsf{p} : \mathbb{N} \to \mathbb{N}$*, there are a strictly increasing* q *and a multilinear table* S *such that the Wilson family converges to* S *along* $\mathsf{v} = \mathsf{p} \circ \mathsf{q}$ *in the sense of Definition 4.3:*

$$\lim_{k\to\infty} \lim_{N\to\infty} C^{\mathrm{lat}}_{N,\mathsf{v}(k)}(v;f) = \int f(x) S_n(v;x)\, dx \,.$$

The table satisfies E1–E7.

The proof is in Section 14, II.F.1–F.3. The extraction first gives volume limits on Gen_n; the difference bounds extend them smoothly to Conf_n. Uniqueness identifies each extracted fixed-volume kernel with $f_{p,v}$, retaining UV's full inner limit at every admissible volume. Reflection positivity passes from Wilson expectations to finite sums and then to joint smeared words. Fix this refinement and its table S from now on.

Write $v(x) = (v_1(x_1), \ldots, v_n(x_n))$ for a spectator word. Two polynomial labels represent the same observable when no separated correlation distinguishes them.

Definition 4.8 (Observable fields)**.** Set

$$\begin{aligned} \mathcal{N}_d(S) &= \big\{ c \in V_d : S_{n+1}(c(x_0), v(x)) = 0 \\ &\qquad\qquad \text{for every } n, v \text{ and } x \in \mathrm{Conf}_{n+1} \big\}, \\ F_d &= V_d/\mathcal{N}_d(S), \qquad \overline{[c]} = [\bar{c}], \\ \iota_d[c](v;x) &= S_{n+1}(c(x_0), v(x)), \qquad \mathcal{L} = \mathrm{span} \bigcup_d \iota_d F_d \,. \end{aligned}$$

The maps ι_d give the responses to all spectators. Choose conjugation-invariant quotient norms.

Permutation symmetry makes correlations independent of representatives in every slot. The unit field is $\mathbf{1}_d = [a^d_{m_{\mathrm{ref}}(d)} \mathbf{1}]$, with the normalization of Definition 3.8.

To state the dimension of F_d, introduce the free comparison space.

Definition 4.9 (Free Maxwell comparison)**.** For $d_{\mathfrak g}$ independent colours, take

$$\mathbb{E}A_a^\alpha(x)A_b^\beta(y) = \delta_{\alpha\beta}\delta_{ab}\Gamma(x-y), \qquad \Gamma(x) = \frac{1}{4\pi^2|x|^2}, \qquad F_{ab} = \partial_a A_b - \partial_b A_a\,.$$

Let M_d^{Max} consist of $\mathrm{Ad}(G)$-invariant Wick polynomials in curvature jets of weight at most d, modulo antisymmetry, the linear Bianchi identities and $\sum_a \partial_a F_{ab} = 0$. Their separated Wick correlations are S^{Max}.

The quotient fields admit inclusions and differentiation that preserve their correlations, and have the Maxwell dimensions.

Proposition 4.10 (Field calculus)**.** *For all $d \le e$, there are injective compatible inclusions $\jmath_{d,e} : F_d \to F_e$ and linear maps $\partial_a : F_d \to F_{d+1}$, with*

$$\dim F_d = \dim M_d^{\mathrm{Max}}, \qquad \mathcal{L} = \bigcup_d \iota_d F_d,$$
$$\partial_{x_i^a} S_n(A;x) = S_n(A_1,\dots,\partial_a A_i,\dots,A_n;x)\,.$$

The proof is in Section 14, II.F.2. It constructs the corrected coefficient maps on the refinement already chosen for IV, preserves units and correlations, and proves compatibility with conjugation and the discrete symmetries. Every quotient field has a token representative.

5 Stress and rotations

We compare the fields of Section 4 with Maxwell fields at small distances, then normalize a conserved tensor by its shell fluxes. The volume refinement and the table S remain fixed throughout. Subsequences taken below concern only comparison radii.

The comparison permits the basis of local fields to vary with the radius. This accommodates mixing among fields in a fixed finite-dimensional space.

Definition 5.1 (Scaling frames)**.** A scaling sequence consists of $r_k > 0$, $r_k \to 0$, and isomorphisms

$$\mathsf{U}_{k,d}^{\mathrm{sc}} : M_d^{\mathrm{Max}} \longrightarrow F_d\,.$$

It is stable if, at every fixed degree horizon, these maps eventually preserve conjugation, H_4, units and inclusions exactly. Scaling dimensions are included in the maps: an ordinary comparison insertion is $\mathsf{U}_{k,d}^{\mathrm{sc}}u$ at $r_k x$, with no further factor r_k^d.

The first comparison concerns correlations integrated against joint tests. A single sequence works for all fixed lists of fields.

Proposition 5.2 (Maxwell comparison)**.** *There is one stable scaling sequence such that, for every $n, u_i \in M_{d_i}^{\mathrm{Max}}$, compact $\mathcal{K} \subset \mathrm{Conf}_n$, $M \ge 0$ and $\varepsilon > 0$, some $k_0(n,u,\mathcal{K},M,\varepsilon)$ gives, for $k \ge k_0$ and $f \in C_c^\infty$ supported in $\mathcal{K}$ with $\|f\|_\infty \le M$,*

$$\left| \int f(x) \big[S_n(\mathsf{U}_{k,d_i}^{\mathrm{sc}} u_i; r_k x) - S_n^{\mathrm{Max}}(u;x) \big]\, dx \right| \le \varepsilon M\,.$$

The proof is in Section 15, II.G.2.1. The lattice comparison stops near the requested physical scale and compares the resulting correlations with the free Gaussian ones. Finitely many separating correlations make the comparison maps injective; the dimension equality of FC makes them isomorphisms. Finite-dimensional corrections give the exact eventual compatibilities in D12.

To read translation and angular charge, integrate a conserved tensor through a shell surrounding just one insertion. Let $A \in F_d$, let $\boldsymbol{B}$ be a spectator field list at x, and take $(x_0, x) \in \mathrm{Conf}_{n+1}$.

Definition 5.3 (Shell fluxes)**.** For $J_{ab} \in F_4$ and $K_{\mathrm{rot}} \in \mathfrak{so}(4)$, choose real $\chi \in C_c^\infty$ equal to one near x_0 and supported away from spectators. Define

$$\mathcal{P}_b^J = -\sum_a \int \partial_a \chi(y) S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\, dy,$$
$$\mathcal{Y}_{K_{\mathrm{rot}}}^J = -\sum_{a,b} \int \partial_a \chi(y)(K_{\mathrm{rot}}(y - x_0))_b S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\, dy\,.$$

These fluxes avoid every insertion point.

The centered real symmetric H_4-tensorial conserved tensors of degree four form a real line, as proved in II.G.2.2. For a nonzero member J, II.G.2.3 constructs linear charge maps $\mathsf{p}_b : F_d \to F_{d+1}$ and $\mathsf{Y}_{K_{\mathrm{rot}}} : F_d \to F_d$ by

$$\mathcal{P}_b^J = S((\mathsf{p}_b A)(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_{K_{\mathrm{rot}}}^J = S((\mathsf{Y}_{K_{\mathrm{rot}}} A)(x_0), \boldsymbol{B}(x))\,.$$

Here centered means that every component has zero one-point function. The charge maps are complex linear in A, and the angular map is real linear in K_{rot}.

Section 15, II.G.3.1, identifies these maps with differentiation and rotations. It constructs continuous complex linear representations R_d of $O(4)$, extending H_4, fixing units and commuting with conjugation and inclusions. For a real $c_\star \neq 0$ common to every degree, the generators and charge maps satisfy

$$\mathsf{L}_d(K_{\mathrm{rot}}) = \frac{d}{dt} R_d(e^{tK_{\mathrm{rot}}})\Big|_{t=0}, \qquad \mathsf{p}_b = c_\star \partial_b, \qquad \mathsf{Y}_{K_{\mathrm{rot}}} = c_\star \mathsf{L}_d(K_{\mathrm{rot}})\,.$$

The proof controls exterior angular flux before extending the rotation identity to all separated configurations.

This common constant fixes the normalization of the stress tensor by its action on fields.

Theorem 5.4 (Stress tensor)**.** *There is a real symmetric $O(4)$-tensor $T_{ab} \in F_4$ with $\sum_a \partial_a T_{ab} = 0$ such that, for every degree, field, spectator list, separated configuration and cutoff of Definition 5.3, every b and every $K_{\mathrm{rot}} \in \mathfrak{so}(4)$,*

$$\mathcal{P}_b^T = \partial_{x_0^b} S(A(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_{K_{\mathrm{rot}}}^T = S((\mathsf{L}_d(K_{\mathrm{rot}})A)(x_0), \boldsymbol{B}(x))\,.$$

The proof is in Section 15, II.G.3.1–G.3.2. Divide the raw current by $c_\star$ and average its tensor transform over $O(4)$ with probability Haar measure; its shell actions are preserved. The table has full covariance,

$$S_n(R_{d_i}(h)A_i; hx) = S_n(A; x), \qquad R_4(h)T_{ab} = \sum_{i,j} h_{ia} h_{jb} T_{ij} \quad (h \in O(4))\,.$$

The free stress T^{Max} is normalized by the same shell laws in II.G.1. In particular their sign gives

$$T^{\mathrm{Max}}_{ab} = -\sum_{c,\alpha} : F^{\alpha}_{ac} F^{\alpha}_{bc} : + \frac{\delta_{ab}}{4} \sum_{i,j,\alpha} : F^{\alpha}_{ij} F^{\alpha}_{ij} : \, .$$

The final comparison includes any chosen stress insertions. For a free list $u_i \in M^{\mathrm{Max}}_{d_i}$, choose a subset I of slots and component pairs (a_i, b_i) on I, and put

$$A^0_i = \begin{cases} T^{\mathrm{Max}}_{a_i b_i}, & i \in I, \\ u_i, & i \notin I. \end{cases}$$

The normalized tensor then has the free limit together with all fixed positional derivatives.

Proposition 5.5 (Differentiated short-distance comparison)**.** *There is one stable scaling sequence such that, for every fixed list and stress-slot choice, every derivative order j and compact $\mathcal{K} \subset \mathrm{Conf}_n$, with $A_{k,i} = r_k^4 T_{a_i b_i}$ on I and $A_{k,i} = \mathsf{U}^{\mathrm{sc}}_{k,d_i} u_i$ elsewhere,*

$$D^j_x S_n(A_k; r_k x) \longrightarrow D^j_x S^{\mathrm{Max}}_n(A^0; x) \quad \textit{uniformly on } \mathcal{K}\,.$$

The proof is in Section 15, II.G.3.3. Write $\partial^{\mathrm{Max}}_a$ for free field differentiation and $\mathsf{L}^{\mathrm{Max}}_d$ for the generators of the natural rotation action on Wick polynomials. Along this same sequence, for every fixed d, a, K_{rot}, the transported infinitesimal actions satisfy the operator-norm limits

$$r_k (\mathsf{U}^{\mathrm{sc}}_{k,d+1})^{-1} \partial_a \mathsf{U}^{\mathrm{sc}}_{k,d} \longrightarrow \partial^{\mathrm{Max}}_a,$$
$$(\mathsf{U}^{\mathrm{sc}}_{k,d})^{-1} \mathsf{L}_d(K_{\mathrm{rot}}) \mathsf{U}^{\mathrm{sc}}_{k,d} \longrightarrow \mathsf{L}^{\mathrm{Max}}_d(K_{\mathrm{rot}})\,.$$

In the correlation limit D^j_x differentiates the whole function of x, including the rescaled positions $r_k x$.

6 Phases and reconstruction

Keep the table and quotient fields of Sections 4–5. Compression to zero energy separates the mixed correlations into phases.

Definition 6.1 (OS space and compression)**.** For words at distinct points in $x^a > 0$, let $\mathcal{H}^{\mathrm{OS}}_a$ be the completion of their complex span modulo the null space of $\langle w, v \rangle_a = S(\mathsf{s}^{\mathrm{refl}}_a w^* v)$. Write Ω^{OS}_a for the empty word and P^0_a for projection onto $\mathcal{H}^0_a = \ker H^{\mathrm{OS}}_a$, with $e^{-tH^{\mathrm{OS}}_a}[v] = [\tau_{te_a} v]$ for $t \geq 0$. For $h_{\mathrm{sl}} > 0$ and a word w with times in $(0, h_{\mathrm{sl}})$, set

$$A_w v = w\, \tau_{h_{\mathrm{sl}} e_4} v, \qquad M^0(w) = P^0_4 \mathsf{B}_{h_{\mathrm{sl}}}(w)|_{\ker H^{\mathrm{OS}}_4}\,,$$

where $\mathsf{B}_{h_{\mathrm{sl}}}(w)$ is the bounded extension proved in Section 16.

Translate any separated word into a slab to define its compression; H.1.2 in Section 16 proves independence of width and placement.

The compressed operators generate the separable commutative unital C*-algebra $\mathfrak{A}_0 = C^*(M^0(w) : w \text{ separated})$. Its characters give the phase correlations.

Theorem 6.2 (Phase decomposition)**.** *The compression algebra has a compact metric character space X with a full-support probability measure ν such that, for every separated word w,*

$$S^{\xi}(w) = \xi(M^0(w)), \qquad S(w) = \int_{\mathsf{X}} S^{\xi}(w)\, d\nu(\xi)\,.$$

Each phase satisfies E1–E7 on the same quotient fields, with the same field calculus, rotations and stress laws. For fixed w, its value is continuous in ξ.

The proof is in Section 16, H.1.1–H.1.3. Commutativity follows by separating four words and exchanging the middle two using Euclidean rotations. The empty-word vector is cyclic and separating for $\mathfrak{A}_0$. Fix ξ_* in the full-measure set with one-dimensional time-axis OS ground space supplied by H.1.3.

Write $\mathcal{S}_n = \mathcal{S}(\mathbb{R}^{4n};\mathbb{C})$ and $\mathcal{S} = \mathcal{S}_1$. The following definitions specify the relativistic theory.

Definition 6.3 (Relativistic fields)**.** Take a complex Hilbert space $\mathcal{H}$, a complex label space $\mathcal{L}$, a linear subspace $\mathcal{D} \subset \mathcal{H}$ and $\Omega \in \mathcal{D}$. Fields $\Phi(\ell, f) : \mathcal{D} \to \mathcal{D}$ are linear operators, complex bilinear in $(\ell, f) \in \mathcal{L} \times \mathcal{S}$. Include an antilinear map $\ell \mapsto \ell^*$, strongly continuous unitary translations $\mathsf{T}(z)$, $z \in \mathbb{R}^4$, and a Borel projection-valued measure E, with finite diagonal measures $\mu_\psi(B) = \langle \psi, \mathsf{E}(B)\psi\rangle$ for Borel $B \subset \mathbb{R}^4$.

Definition 6.4 (Poincaré action)**.** Take a topological group $\mathcal{P}$ with group homeomorphism $\kappa_{\mathrm{P}} : \mathcal{P} \to SO^+(1,3) \ltimes \mathbb{R}^4$, writing $\kappa_{\mathrm{P}}(\gamma) = (\Lambda_\gamma, z_\gamma)$. Here $SO^+(1,3)$ preserves m, has determinant one and preserves $\overline{V}_+$. Include a strongly continuous unitary representation U with $\mathsf{U}(1,z) = \mathsf{T}(z)$ and linear bijections $\rho_{\mathrm{lab}}(\gamma)$ of $\mathcal{L}$. Set $(\gamma \cdot f)(x) = f(\Lambda_\gamma^{-1}(x - z_\gamma))$.

The antilinear label map need not be involutive. The label action has no additional continuity, dimension or translation restriction. The spectral relation is imposed next.

Definition 6.5 (W0. Spectrum and vacuum)**.** For every $\psi \in \mathcal{H}$ and $z \in \mathbb{R}^4$, require

$$\langle \psi, \mathsf{T}(z)\psi\rangle = \int e^{i\mathbf{p}\cdot z}\, d\mu_\psi(\mathbf{p}), \qquad \mathsf{E}(\mathbb{R}^4 \setminus \overline{V}_+) = 0\,,$$

with Euclidean pairing. Every vector fixed by all translations belongs to $\mathbb{C}\Omega$.

Definition 6.6 (W1. Word domain and adjoints)**.** Require $\|\Omega\| = 1$, density of $\mathcal{D}$, and

$$\mathcal{D} = \mathrm{span}_{\mathbb{C}}\{\Phi(\ell_1, f_1)\cdots\Phi(\ell_k, f_k)\Omega : k \geq 0\}\,.$$

For each $\ell \in \mathcal{L}$, $\psi \in \mathcal{D}$, the map $f \mapsto \Phi(\ell, f)\psi$ is continuous from $\mathcal{S}$ to $\mathcal{H}$ in norm. For $\psi, \chi \in \mathcal{D}$, require $\langle \Phi(\ell, f)\psi, \chi\rangle = \langle \psi, \Phi(\ell^*, \bar{f})\chi\rangle$.

Definition 6.7 (W2. Covariance)**.** Require $\rho_{\mathrm{lab}}(1) = 1$ and $\rho_{\mathrm{lab}}(\gamma\gamma') = \rho_{\mathrm{lab}}(\gamma)\rho_{\mathrm{lab}}(\gamma')$. For every $\gamma \in \mathcal{P}$, $\ell \in \mathcal{L}$, $f \in \mathcal{S}$ and $\psi \in \mathcal{D}$, require $\mathsf{U}(\gamma)\mathcal{D} \subset \mathcal{D}$, $\mathsf{U}(\gamma)\Omega = \Omega$, and

$$\mathsf{U}(\gamma)\Phi(\ell, f)\psi = \Phi(\rho_{\mathrm{lab}}(\gamma)\ell, \gamma \cdot f)\mathsf{U}(\gamma)\psi\,.$$

Definition 6.8 (W3. Locality)**.** For $f, h \in \mathcal{S}$ with $\mathsf{m}(x - y) < 0$ for every $x \in \operatorname{supp} f$, $y \in \operatorname{supp} h$, require $[\Phi(\ell, f), \Phi(\ell', h)]\psi = 0$ for all $\ell, \ell' \in \mathcal{L}$ and $\psi \in \mathcal{D}$.

Definition 6.9 (Wightman axioms)**.** A relativistic datum of Definitions 6.3–6.4 is a Wightman theory if it satisfies W0–W3.

Definition 6.10 (Faithful labels)**.** The labels are faithful if, for every nonzero $\ell \in \mathcal{L}$, some $f \in \mathcal{S}$ has $\Phi(\ell, f) \neq 0$.

Definition 6.11 (Bounded weak commutant)**.** A bounded operator B belongs to the weak commutant if, for every smeared field Φ and all $u, v \in \mathcal{D}$, one has $u \in \operatorname{Dom}\Phi^*$ and $\langle \Phi^* u, Bv\rangle = \langle u, B\Phi v\rangle$. Here Φ^* is the unbounded Hilbert-space adjoint.

Definition 6.12 (Hermitian scalar labels)**.** A label is Hermitian if $\ell^* = \ell$, and Lorentz scalar if $\rho_{\mathrm{lab}}(\Lambda, 0)\ell = \ell$ for every $\Lambda \in SO^+(1,3)$. These conditions give the adjoint and scalar covariance identities through W1 and W2.

For reconstruction use the response space $\mathcal{L}$ of Definition 4.8. Put $\mathbb{F} = \bigsqcup_d F_d$ and $\iota(d, A) = \iota_d A$.

Definition 6.13 (Boundary condition)**.** For a phase ξ and functions F_n^ξ on $\mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n$, require, for every $n \in \mathbb{N}$, $\boldsymbol{A} \in \mathbb{F}^n$, real $h_1 < \cdots < h_n$ and $f_i \in \mathcal{S}$,

$$\lim_{r \downarrow 0} \int \mathsf{F}_n^\xi(\boldsymbol{A}; (x_i^4 + irh_i)_i, (\vec{x}_i)_i) \prod_i f_i(x_i)\, dx \\ = \langle \Omega_\xi, \Phi_\xi(\iota(A_1), f_1) \cdots \Phi_\xi(\iota(A_n), f_n)\Omega_\xi \rangle \,.$$

For this formula alone a nonintegrable function has integral zero.

Definition 6.14 (Ordered-time continuation)**.** Put $\mathcal{Z}_n = \{z \in \mathbb{C}^n : \operatorname{Im} z_1 < \cdots < \operatorname{Im} z_n\}$. A continuation is holomorphic there at each fixed real spatial configuration and satisfies

$$\mathsf{F}_n^\xi(\boldsymbol{A}; (it_i)_i, \vec{x}) = S_n^\xi(\boldsymbol{A}; (\vec{x}_i, t_i)_i) \qquad (t_1 < \cdots < t_n) \,.$$

Together with Definition 6.13, it connects Euclidean and real-time correlations.

Apply Definition 6.1 to S^ξ to obtain $\mathcal{H}_{4,\xi}^{\mathrm{OS}}$ with empty-word vector Ω_ξ^{OS}. The OS comparison map is the bounded map from the reconstructed space obtained by continuing positive-time words. Write $\mathsf{T}_\xi(te_4) = e^{itH_\xi}$ for its energy generator. Each character yields relativistic fields on the common response labels.

Theorem 6.15 (Reconstruction)**.** *For every $\xi \in \mathsf{X}$, the phase correlations have a faithful Wightman reconstruction on $\mathcal{L}$, with the continuation and boundary condition of Definitions 6.14 and 6.13. Its OS comparison map is onto, and*

$$\ker H_\xi = \mathbb{C}\Omega_\xi, \qquad \ker H_{4,\xi}^{\mathrm{OS}} = \mathbb{C}\Omega_\xi^{\mathrm{OS}}, \qquad \{\textit{bounded weak commutant}\} = \mathbb{C}\mathrm{Id}_{\mathcal{H}_\xi} \,.$$

The proof is in Section 16, H.2.1–H.2.4: slab bounds control the analytic boundary values. Those results also give joint Schwartz distributions, uniform low-energy bounds, phase averages and stress continuation. Retain the theory at ξ_* and suppress its phase subscript.

7 Trace and products

The trace is centered in the phase already selected in Section 6. Its short-distance logarithm will distinguish the resulting theory from finite Wick comparisons.

Definition 7.1 (Centered trace)**.** For the tensor of Theorem 5.4, put

$$\tau = \sum_{a=1}^{4} T_{aa}, \qquad e_{\mathrm{tr}} = S^{\xi_*}(\tau), \qquad \Theta = \tau - e_{\mathrm{tr}}\mathbf{1}_4,$$

$$G_\Theta(r) = S_2^{\xi_*}(\Theta, \Theta; -re_4/2, re_4/2), \qquad A_G = \frac{3d_{\mathfrak{g}}}{\pi^4} \,.$$

The one-point value is position independent. Centering removes the vacuum contribution in the selected phase.

The normalized stress has a nonzero centered trace. The next limit determines both its leading power and its logarithmic factor.

Theorem 7.2 (Trace law)**.** *The selected field Θ is nonzero, real and $O(4)$ invariant; its relativistic field is a Hermitian Lorentz scalar. For every $f \in \mathcal{S}$,*

$$\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0, \qquad \lim_{r\downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = A_G > 0\,.$$

The proof is in Section 17, I.1.1–I.1.3. The degree-four comparison combines the scalar coefficient $d_2 = -b_0/2$ with the running-coupling slope $\kappa_G = 2b_0$. Its leading trace coefficient is therefore $4d_2/(\kappa_G \log(1/r)) = -1/\log(1/r)$. Squaring it gives the logarithmic factor; the free curvature-square correlation gives A_G. The comparison uses adjustable stopping levels and fourth-order information beyond Proposition 5.2. It proves the limit along all positive radii.

The same proof identifies centering on the word domain:

$$\Phi(\iota_4\Theta, f) = \Phi(\iota_4\tau, f) - e_{\mathrm{tr}}\left(\int f\right)\mathrm{Id}_{\mathcal{H}}\,.$$

It also constructs the unique positive tempered-growth spectral measure μ_Θ, supported in $\overline{V}_+$, for which

$$d\mu_{\Phi(\iota_4\Theta,f)\Omega} = |\widehat{f}_+|^2\, d\mu_\Theta, \qquad G_\Theta(r) = \int e^{-r\mathbf{p}^4}\, d\mu_\Theta(\mathbf{p}) \quad (r>0)\,.$$

Here $\widehat{f}_+(\mathbf{p}) = \int e^{i\mathbf{p}\cdot x} f(x)\, dx$. The measure's Lorentz invariance and its high-energy limits against compactly supported continuous tests and Schwartz tests are stated in I.1.2–I.1.3.

Definition 7.3 (Finite Wick profiles)**.** For $r > 0$, $m_{\mathrm{sh}} \geq 0$, integers $j_h, k_E \geq 0$ and continuous real h_{mom} on $\mathbb{R}^3$ with $h_{\mathrm{mom}}(\lambda\mathbf{k}) = \lambda^{j_h} h_{\mathrm{mom}}(\mathbf{k})$ for $\lambda > 0$, set

$$\omega_{m_{\mathrm{sh}}}(\mathbf{k}) = \sqrt{m_{\mathrm{sh}}^2 + |\mathbf{k}|^2},$$
$$K_{m_{\mathrm{sh}},k_E,h_{\mathrm{mom}}}(r) = \int_{\mathbb{R}^3} \frac{\omega_{m_{\mathrm{sh}}}(\mathbf{k})^{k_E} h_{\mathrm{mom}}(\mathbf{k})}{2\omega_{m_{\mathrm{sh}}}(\mathbf{k})} e^{-r\omega_{m_{\mathrm{sh}}}(\mathbf{k})}\, d^3\mathbf{k}\,.$$

A finite Wick profile is a finite complex linear combination of finite products of these kernels. A zero denominator at zero momentum gives integrand zero. This is the Euclidean comparison class.

Finite products and sums of these kernels cannot reproduce the inverse logarithm in Theorem 7.2.

Proposition 7.4 (Finite-profile exclusion)**.** *For every finite Wick profile W_{Wick} and every $\delta > 0$, there is $0 < r < \delta$ such that*

$$G_\Theta(r) \neq W_{\mathrm{Wick}}(r)\,.$$

The proof is in Section 17, I.2.1. The mass-shell kernels have expansions to every finite order in integer powers of r and nonnegative powers of $\log r$. Finite products and sums therefore give, to bounded error, a finite sum of monomials $r^a(\log r)^k$ with $a \in \mathbb{Z}$ and $k \in \mathbb{N}$. Multiplication by $r^8 \log^2(1/r)$ makes the terms with $a > -8$ vanish. The remaining nonzero leading term diverges, so any finite limit is zero. The positive limit in Theorem 7.2 therefore excludes equality on every sufficiently small interval. I.2.2–I.2.4 give the polynomial mass-shell and Fock countertests, with their comparison classes defined there.

The product expansion is stated for the Euclidean tables on their common quotient fields. Write $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$, with $S^{\mathrm{mix}} = S$, and retain degree tags by $\mathbb{F} = \bigsqcup_d F_d$, $\deg(d, A) = d$. Choose

finite channel sets Ch_j and real fields $O_{j,\kappa} \in F_j$ forming complex bases of conjugation-invariant complements of F_{j-1} in F_j, with $F_{-1} = 0$. Section 18, J.1.1 constructs these fields and coefficient functions $c_{j,\kappa}$ once, before spectators, states and truncations. The degree-zero vector spans the unit line; it need not equal the unit. For finite cluster and spectator index sets $I_{\mathrm{cl}}, I_{\mathrm{sp}}$, the positions are $x \in (\mathbb{R}^4)^{I_{\mathrm{cl}}}$, $y \in (\mathbb{R}^4)^{I_{\mathrm{sp}}}$, with expansion center $o \in \mathbb{R}^4$.

Definition 7.5 (Product remainder)**.** For finite index sets $I_{\mathrm{cl}}, I_{\mathrm{sp}}$ and $D \in \mathbb{N}$, cluster and spectator lists $\boldsymbol{A} \in \mathbb{F}^{I_{\mathrm{cl}}}$, $\boldsymbol{B} \in \mathbb{F}^{I_{\mathrm{sp}}}$, put $|\boldsymbol{A}|_{\deg} = \sum_i \deg A_i$ and

$$\mathcal{R}_D^{\zeta} = S^{\zeta}(\boldsymbol{A}, x; \boldsymbol{B}, y) - \sum_{j=0}^{D} \sum_{\kappa \in \mathrm{Ch}_j} c_{j,\kappa}(\boldsymbol{A}; x, o) S^{\zeta}(O_{j,\kappa}(o), \boldsymbol{B}(y)) \,.$$

Use a fixed enumeration of $I_{\mathrm{cl}} \sqcup I_{\mathrm{sp}}$ for the concatenated correlations. An $(r, M_{\mathrm{geo}}, \eta)$-cluster satisfies $|x_i - o| \le M_{\mathrm{geo}} r$ and $|x_i - x_{i'}| \ge \eta r$ for distinct slots. Spectators are δ_{sp}-clear if $|o - y_j| \ge \delta_{\mathrm{sp}}$ and $|y_j - y_{j'}| \ge \delta_{\mathrm{sp}}$ for distinct slots. The remainder measures the error through degree D.

One field expansion approximates a cluster in every phase and in the mixture. Its coefficients are independent of all spectators.

Theorem 7.6 (Operator product expansion)**.** *For every finite list $\boldsymbol{A}$, $M_{\mathrm{geo}} \ge 0$, $\eta, \varepsilon > 0$, $j \in \mathbb{N}$ and $\kappa \in \mathrm{Ch}_j$, some $r_0 > 0$ gives, for $0 < r < r_0$ and every $(r, M_{\mathrm{geo}}, \eta)$-cluster,*

$$|c_{j,\kappa}(\boldsymbol{A}; x, o)| \le r^{j - |\boldsymbol{A}|_{\deg} - \varepsilon} \,.$$

For every pair of finite lists $\boldsymbol{A}, \boldsymbol{B}$, $M_{\mathrm{geo}} \ge 0$, $\eta, \delta_{\mathrm{sp}}, \varepsilon > 0$ and $D \in \mathbb{N}$, some $r_1 > 0$ gives, for $0 < r < r_1$, such clusters, δ_{sp}-clear spectators and all states,

$$|\mathcal{R}_D^{\zeta}| \le r^{D+1-|\boldsymbol{A}|_{\deg} - \varepsilon} \,.$$

The proof is in Section 18, J.1.1–J.1.2. Finite correlation functionals first extract a field, rotation and conjugation averages impose the symmetries, and recursive subtraction removes lower degrees. The lattice product estimate controls the discarded part. Its compressed-algebra norm bound holds before character evaluation, which gives the same radius threshold for every state. The two thresholds have separate dependencies, and each displayed bound has coefficient one.

J.1.3–J.1.6 give integrated and nested estimates; the nested error is a sum of gains from proper clusters.

Separated product estimates leave the collision diagonals untreated. Contact subtractions use the same underlying prescription, with their own projections, separating correlation functionals and clearance.

Theorem 7.7 (Contact extension)**.** *For every state ζ, arity n and list $A_i \in F_{d_i}$, the fixed contact construction gives a distribution $\mathfrak{D}_{\boldsymbol{A}}^{\zeta}$ on $\mathbb{R}^{4n}$ such that*

$$\mathfrak{D}_{\boldsymbol{A}}^{\zeta}(f) = \int f(x) S_n^{\zeta}(\boldsymbol{A}; x)\, dx \qquad (f \in C_c^{\infty}(\mathrm{Conf}_n)) \,.$$

The proof is in Section 18, J.2.1–J.2.3. Taylor subtraction in the directions normal to collision diagonals is organized by forests of disjoint or nested subsets. The contact projections retain compatible field components, while the construction's lattice bounds make the shell sums converge. That section specifies the real stratum bases, subtraction scales and local finite-order bounds. These contact estimates are additional to the separated OPE bounds.

8 Main theorems

The existence theorem concerns one lattice family, chosen before the volume sequence. We first specify its continuum conclusion; the field calculus and stress tensor established above are additional properties of the construction.

Definition 8.1 (Hierarchy data)**.** For a Wilson family $\mathcal{W}$ with token set $\mathcal{T}$, hierarchy data consist of:

- complex Hilbert and vector spaces $\mathcal{H}, \mathcal{L}$, and the relativistic data of Definitions 6.3–6.4;
- finite-dimensional complex normed spaces F_d, linear maps $\iota_d : F_d \to \mathcal{L}$, and $\pi_{\mathrm{quo}} : \mathcal{T} \to \mathbb{F}$, where $\mathbb{F} = \bigsqcup_d F_d$;
- a compact metric space X, a Borel measure ν and $\xi_* \in \mathsf{X}$;
- functions $S_n^\zeta : \mathbb{F}^n \times (\mathbb{R}^4)^n \to \mathbb{C}$ for $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$, and $\mathsf{F}_n : \mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n \to \mathbb{C}$;
- finite sets Ch_j, fields $O_{j,\lambda} \in F_j$, and coefficient functions $c_{j,\lambda} : \mathbb{F}^I \times (\mathbb{R}^4)^I \times \mathbb{R}^4 \to \mathbb{C}$ for every finite index set I;
- an element $\Theta \in F_4$.

No further condition is part of these data.

Put $\deg(d, A) = d$ and $\iota(d, A) = \iota_d A$. Here the spaces and maps are arbitrary: no response realization, stress interpretation, multilinearity or channel basis property is implicit. Correlations are defined at all positions; the axioms constrain their separated values.

Definition 8.2 (Regular lattice approximation)**.** A Wilson family is regular if its insertions are continuous and gauge invariant, each token v has a finite stencil $\Sigma_v \subset \mathbb{Z}^4$ valid for every N, p, y, and some $b_* > 0$ satisfies

$$\forall \varepsilon > 0 \ \exists N_0 \ \forall N \geq N_0 \ \forall p : \quad \left| \frac{\beta_{N,p}}{\log(1/a_N)} - b_* \right| < \varepsilon \,.$$

Only links with both ends in $y + \Sigma_v$ modulo the torus are read. The continuum approximation is Definition 4.3 with $K_n(v; x) = S_n^{\mathrm{mix}}(\pi_{\mathrm{quo}}(v_1), \dots, \pi_{\mathrm{quo}}(v_n); x)$.

The relativistic axioms Q2 are Definition 6.9. The following conditions specify the labels, phase average and selected continuation.

Definition 8.3 (Labels and phases)**.** Every nonzero $\ell \in \mathcal{L}$ has $\Phi(\ell, f) \neq 0$ for some $f \in \mathcal{S}$. Each ι_d is injective, their images cover $\mathcal{L}$, π_{quo} is onto, and some $\Phi(\iota(a), f)$ is nonzero. The measure ν is a probability measure of full support. Each S_n^ζ is simultaneously permutation symmetric and smooth on Conf_n. For every fixed separated field configuration,

$$\xi \longmapsto S_n^\xi(a; x) \text{ is continuous}, \qquad S_n^{\mathrm{mix}}(a; x) = \int_{\mathsf{X}} S_n^\xi(a; x) \, d\nu(\xi) \,.$$

Definition 8.4 (Selected continuation)**.** The functions F_n satisfy the ordered-time continuation condition of Definition 6.14 and the boundary condition of Definition 6.13 at ξ_*. In those definitions write $\mathsf{F}_n^{\xi_*} = \mathsf{F}_n$.

Definition 8.5 (Products)**.** The single coefficient system of Definition 7.5 satisfies both inequalities of Theorem 7.6 with its quantifier order: the coefficient threshold depends on $I, a, M, \eta, \varepsilon, j, \lambda$; the remainder threshold depends on $I, J, a, B, M, \eta, \delta, \varepsilon, D$. Both precede positions and states. Coefficients have no spectator, state or truncation argument.

Definition 8.6 (Trace test)**.** For the chosen $\Theta \in F_4$, define G_Θ and A_G by the last two formulas of Definition 7.1. Its vacuum expectation vanishes for every Schwartz smearing, $A_G > 0$, the full-radius limit of Theorem 7.2 holds in $\mathbb{C}$, and the finite-profile exclusion of Proposition 7.4 holds.

These conditions state the exact existence conclusion. In particular, Q6 places a condition on a degree-four field without requiring a stress-tensor interpretation.

Theorem A (Existence)**.** *For every gauge group of Definition 2.1, there is a regular* (Q1) *Wilson family $\mathcal{W}$ such that, for every strictly increasing $\mathsf{p} : \mathbb{N} \to \mathbb{N}$, there are a strictly increasing q and hierarchy data over $\mathcal{W}$* (Q0) *satisfying* Q2–Q6*, with*

$$\mathcal{W} \longrightarrow S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}} \quad \textit{along } \mathsf{p} \circ \mathsf{q} \quad \textit{in the sense of Definition 4.3}.$$

The deduction below proves Theorem A from the construction theorem. The latter retains the field calculus and quantitative estimates; subsidiary results are identified by their letter and number in Sections 9–18.

Theorem B (Construction)**.** *For every gauge group of Definition 2.1 and $\mathsf{b} > 0$, there is one regular Wilson family $\mathcal{W}$, given by Definitions 3.1 and 3.8 along MT, satisfying* E.1.1–E.1.3, A.1, E.2.1–E.2.1a (B1)*, and* UV, E.3.1–E.3.2 (B2)*. For every strictly increasing p, there are a strictly increasing q, the table S of* IV *and hierarchy data over $\mathcal{W}$ satisfying* Q2–Q6*, with $\mathcal{W} \to S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}}$ along $\mathsf{p} \circ \mathsf{q}$* (Definition 4.3)*. The fields $F_d = V_d/\mathcal{N}_d(S)$ and tensor* ST *obey* FC, F.2–F.3, MC, SC, G.2.1–G.3.3, H.1.3 (B3)*; its phases* PH *obey* RC, H.2.1–H.2.4 (B4)*; its centered trace* (Definition 7.1) *obeys* TR, NF, I.1.1–I.2.4 (B5)*; its prescription* J.1.1 *obeys* OP, J.1.1–J.1.6, CT, J.2.1–J.2.3 (B6).

The proof follows in ten steps. The degree-four position normalization and general collision-subspace estimates remain at their proof locations outside these conclusions. Restriction of the full ultraviolet limit also gives the spacing-subsequence version.

Deduction of Theorems A and B.

1. Fix G and $\mathsf{b} > 0$. Appendix A and Sections 9–13 supply the common choices, admissible classes, recursion and later lattice estimates. Wilson initialization D.3 starts the induction; Theorem 3.5 advances it; Theorem 3.6 and E.1.3 tune trajectories to the common endpoint. These choices precede any volume request.

2. Fix the local sections, reference levels and terminal maps on those trajectories. Definition 3.8 defines every insertion, including below its reference level. Polynomial stencil dependence gives continuity, gauge invariance and tokenwise locality. Taking $m = N$ in Theorem 3.6 gives Q1 with $b_* = \kappa_G > 0$. This fixes the regular Wilson family.

3. Proposition 4.1 and fixed-volume comparison with analytic tuning give Theorem 4.2 and E.3.1–E.3.2, for every volume and fixed field/difference request. Thus each admissible test has a full inner limit before volume extraction.

4. Given any strictly increasing p, Theorem 4.7 chooses q, S, including the refinement for corrected coefficient maps. In Section 14, F.1 identifies the extracted volume kernels with the full kernels by uniqueness of limits. Since $L_{\mathsf{p}(\mathsf{q}(k))} \to \infty$, every compact support eventually fits. This proves the approximation of Definition 4.3, without a spacing refinement.

5. Define $F_d = V_d/\mathcal{N}_d(S)$ and $\mathcal{L}$ by Definition 4.8. Proposition 4.10 and F.2 give inclusions, derivatives, dimensions and the response realization. Proposition 5.2 and the current lemmas of Section 15 give the conserved line; Theorem 5.4 gives its normalized tensor and rotations; Proposition 5.5 gives the differentiated comparison. The volume refinement is unchanged.

6. Theorem 6.2 and H.1.1–H.1.3 of Section 16 construct X, ν, S^{ξ} on the same quotient spaces. Choose ξ_* from the full-measure set with one-dimensional time-axis OS ground space. Theorem 6.15 reconstructs every character simultaneously; H.2.1–H.2.4 retain this selected theory and give B4.

7. Set $\pi_{\mathrm{quo}}(d, \mathbf{c}) = (d, [\mathbf{c}])$. Representatives make this map onto. Every response lies in one $\iota_d F_d$, and each ι_d is injective. The unit response is nonzero since its one-point function is one; faithfulness gives a nonzero acting field. The Euclidean symmetry and smoothness conditions and Theorem 6.2 give Q3; reconstruction gives Q2 and Q4. These identifications use the all-spectator quotient.

8. Center the trace of this tensor in this selected phase. Theorem 7.2 and I.1.1–I.1.3 of Section 17 give its operator identity, full-radius asymptotic and unique spectral measure. Proposition 7.4 gives Q6's finite-profile exclusion; I.2.2–I.2.4 give the additional countertests in B5, with ξ_* unchanged.

9. Choose the prescription of J.1.1 once. Theorem 7.6 gives Q5; J.1.3–J.1.6 give the integrated and nested bounds in B6. Theorem 7.7 and J.2.1–J.2.3 of Section 18 use the same underlying prescription, with separate contact projections, functionals and clearance, to give the extensions, real bases and subtraction scales. This uses its own lattice estimates.

10. Steps 1–9 give B1–B6 on the same family, table, quotient fields, tensor, phases and selected theory. Retain Q1–Q6 and set $\mathsf{b} = 1$. Their quantifier order, lattice approximation, reconstruction, products and trace test give Theorem A.

□

Part II
Proofs

9 Fields, labels and norms

This section is under construction.

10 The renormalization step

This section is under construction.

Definition 10.1 (Axial frames)**.** This definition will be stated here; the section is under construction.

Definition 10.2 (Coarse predictors)**.** This definition will be stated here; the section is under construction.

11 Effective actions and the marginal flow

This section is under construction.

Definition 11.1 (Curvature slots)**.** This definition will be stated here; the section is under construction.

Definition 11.2 (Path action)**.** This definition will be stated here; the section is under construction.

12 The nonlinear step

This section is under construction.

Definition 12.1 (Regular activities)**.** This definition will be stated here; the section is under construction.

Definition 12.2 (Forced activities)**.** This definition will be stated here; the section is under construction.

Definition 12.3 (Continuity of representations)**.** This definition will be stated here; the section is under construction.

13 The multiscale construction

This section is under construction.

14 Infinite volume

This section is under construction.

15 The Maxwell comparison, Ward identities and the stress tensor

This section is under construction.

16 Phases and reconstruction

This section is under construction.

17 The trace law and non-freeness

This section is under construction.

18 The operator product expansion and contact extensions

This section is under construction.

A Constants and order of choice

Table 1 gives the choices for the construction. At each fixed stage the requirements in Sections 10–13, 15.2, 17.1 and 18 impose finitely many lower bounds, depending on earlier data. Costs determining a ratio precede that ratio; subsequent entry levels can be increased. The displayed inequalities are properties of these choices.

Table 1: Choices in dependence order. Stage choices are made at each fixed $j \geq 4$ before entry levels.

	Choice	Dependence, restrictions and purpose
1	Gauge normalization; $\mathsf{b} > 0$	Fix G, its realization, probability Haar measure and metric, then $d_{\mathfrak{g}}, C_{\mathfrak{g}}, b_0, \kappa_G$ as in Definition 2.1; $\kappa_G > 0$. All later choices use these data.
2	Frame and predictor cutoffs; axial factor	Depend on G. Choose the cutoffs small enough and $a_{\mathrm{ax}} \geq 16$ so that the branches are valid on every patch (Section 10.1).
3	Homogeneous bases, norms, sections, stencils	Depend on G, d. Impose Taylor matching, reality and H_4 covariance (Section 9.1 and Definitions 3.7–3.8).
4	Stage parameters	Depend on rows 1–3 and j. Choose $\mathfrak{q}_j \geq 6$, $\varpi_j \geq 2$, $\sigma_j = 100$, $w_j^{\mathrm{path}}, A_j^{\mathrm{path}}, R_j^{\mathrm{rel}} > 0$, $C_j \geq 1$, $\omega_j^{\mathrm{RG}} \geq 0$, and clipping constants. Require $R_j^{\mathrm{rel}} w_j^{\mathrm{path}} \geq 12$; the drift exceeds the one-loop residual. Orders and powers tend to infinity and, along levels, are nondecreasing, as are support exponents. All ratio costs are now fixed.
5	Odd ratios ℓ_j^{st}	Depend on rows 1–4; choose $\ell_j^{\mathrm{st}} \geq 3$ nondecreasing and above every ratio threshold. At $j = J_{m+1}$ and $\ell_m = \ell_j^{\mathrm{st}}$ require $\dfrac{\omega_j^{\mathrm{RG}}}{\log \ell_m} \leq \min(\kappa_G/4, j^{-2})$, $\dfrac{\log(1 + A_j^{\mathrm{path}} + (w_j^{\mathrm{path}})^{-1} + C_j + R_j^{\mathrm{rel}} + \sigma_j)}{\log \ell_m} \leq j^{-2}$. These bounds control the drift and logarithmic stage costs relative to $\log \ell_m$.
6	Exponents and coupling entry bounds	Depend on rows 1–5. Choose positive antitone θ_j and bounds β_j^{ent} meeting the finite maximum of the remaining stage requirements, including those of Sections 15.2, 17.1 and 18. Entry bounds may be raised.
7	$\bar{\beta}$, entry levels m_j^{ent}, minimum periods n_j^{min}	Depend on rows 1–6. Require $\bar{\beta} > 1$, coupling and level entry conditions, and $t_m \leq 1$. Delaying entry preserves earlier requirements.
8	$J_m, \ell_m, D_m, a_m, L_p$, $m_{\mathrm{ref}}(d)$	Depend on rows 1–7. Set $D_0 = 1$, $D_{m+1} = \ell_m D_m$, $a_m = D_m^{-1}$, $s_m = \log D_m$, with $\ell_m = \ell_{J_{m+1}}^{\mathrm{st}}$ and $4 \leq J_m \leq J_{m+1} \leq J_m + 1$, $J_m \to \infty$. Require $0 < \theta_{J_m} \leq J_m^{-1}$, $\log \ell_m \to \infty$, $\log \ell_m / s_m \to 0$. Choose strictly increasing L_p with $4 \mid L_p$ and every positive integer dividing infinitely many L_p. Require $L_p D_m \geq 100(B_m + 1)$ and $L_p D_{m+1} \geq 100((J_{m+1} + 3)\ell_m + 1)$. Choose $J_{m_{\mathrm{ref}}(d)} \geq d + 3$, with $\mathfrak{q}_{J_m} \geq d + 1$, $\varpi_{J_m} \geq d$ for $m \geq m_{\mathrm{ref}}(d)$.

Table 1 (continued)

	Choice	Dependence, restrictions and purpose
9	Derived level scales	No new choices. Set $$\widehat{\beta}_m = \bar{\beta} + \kappa_G s_m > 1, \qquad M_m^{\mathrm{cut}} = \exp((\log\log\widehat{\beta}_m)^2)\,,$$ $$t_m = \widehat{\beta}_m^{-1/2} M_m^{\mathrm{cut}}, \qquad B_m = \lceil (\log\widehat{\beta}_m)^2 \rceil\,,$$ $$\epsilon_m^{\mathrm{reg}} = \widehat{\beta}_m^{-11/10}, \qquad \epsilon_m^{\mathrm{for}} = e^{-(M_m^{\mathrm{cut}})^{1/5}}\,,$$ $$\Lambda_m^{\mathrm{loop}} = 2(M_m^{\mathrm{cut}})^{\theta_{J_m}}, \qquad \mathfrak{B}_m = [\widehat{\beta}_m/2, 2\widehat{\beta}_m]\,.$$ These fix the activity bounds.
10	Actions, classes, recursion, trajectories, couplings; width w	Depend on rows 1–9. Choose the path actions and admissible subsets, the specified recursion and its trajectories for every finite N, p, all with the common level-zero endpoint. The width $w > 0$ is independent of N, p (Theorem 3.6, Section 13.1).
11	Marked transfers, terminal maps, insertions	Depend on rows 1–10; impose reference normalization (Definition 3.8). Fix them before any prescribed volume sequence.
12	p, then $\mathsf{q}, S, F_d, \partial_a, \mathcal{L}$	Prescribe arbitrary increasing p first. Theorem 4.7 and Proposition 4.10 give the refinement and continuum fields. Theorem 4.2 retains the full inner limit.
13	Comparison radii, frames, J, charges, $c_\star, R, T$	Depend on row 12 and the fixed lattice family. Use Proposition 5.2, then Theorem 5.4 and Proposition 5.5. Refine only comparison radii.
14	Slab operators, X, ν, S^ξ, then ξ_*	Depend on earlier data. Theorem 6.2 selects ξ_* solely for a one-dimensional OS ground space.
15	Every-phase theories and continuations; Θ, μ_Θ	Depend on earlier data. Theorem 6.15 reconstructs the theories; Theorem 7.2 centers the trace in the already selected phase and supplies its spectral measure.
16	OPE prescription; contact projections, distributions, bases, scales	Depend on rows 12–14 and the fixed family. Theorems 7.6 and 7.7 use one underlying prescription, with separate contact functionals and clearance.
17	Estimate constants and thresholds	Depend only on the fixed data preceding them in each statement. They have no later position, volume or state dependence wherever uniformity is asserted.

References